SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE:CSA1105

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I P.Malreddy (192372015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
P.Malreddy

Annexure A

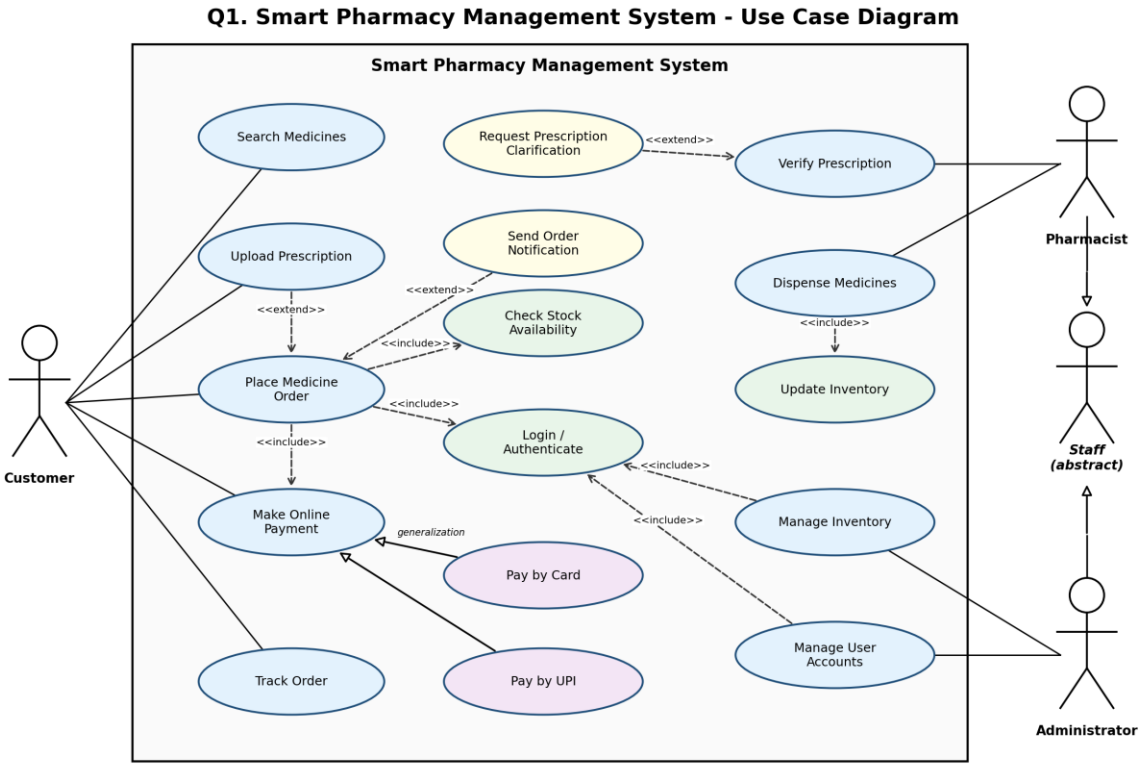
Application Based Questions

1. Use Case Diagram Analysis

A **Smart Pharmacy Management System** allows customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage inventory and user accounts. The current system design does not clearly define actor responsibilities and interactions.

Analyze the system requirements, identify all actors and missing use cases, and design an optimized **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to improve system functionality.

(10 Marks)

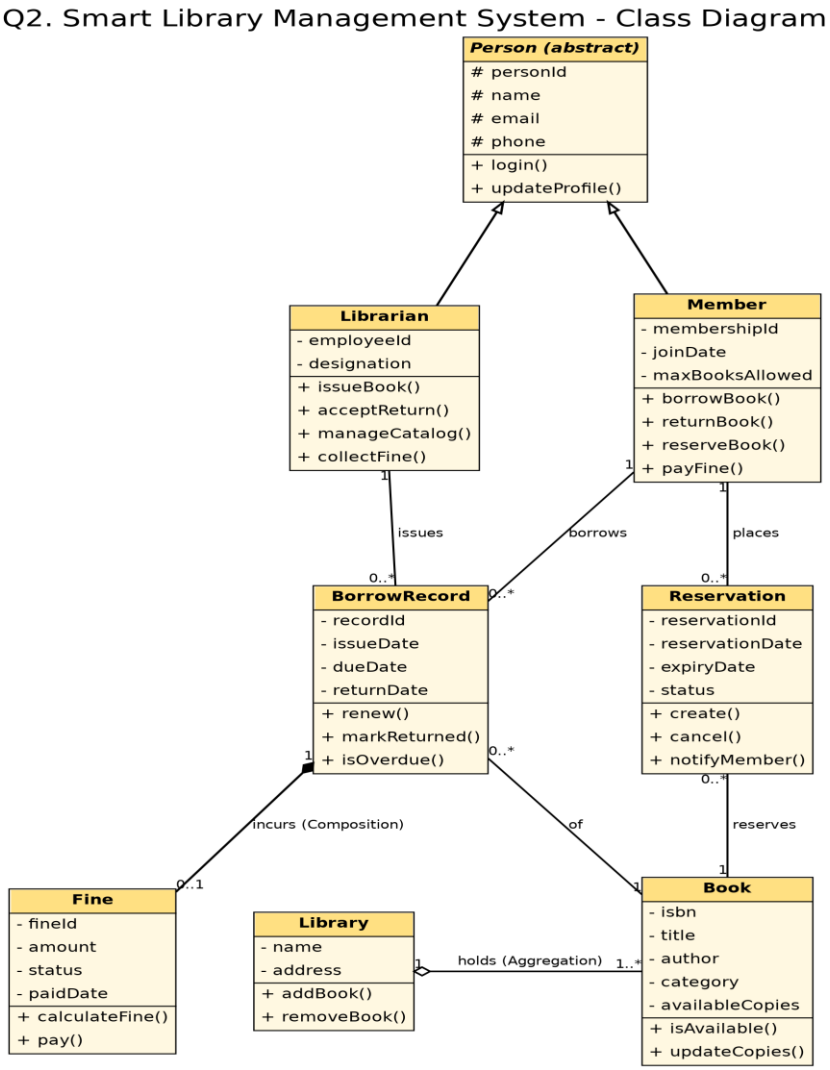


2. Class Diagram Design

A **Smart Library Management System** consists of entities such as Book, Member, Librarian, Reservation, Fine, and BorrowRecord. The existing system does not clearly specify class attributes, operations, or relationships among these entities.

Analyze the system, identify relevant classes with appropriate attributes and methods, define relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a well-structured **Class Diagram**.

(10 Marks)

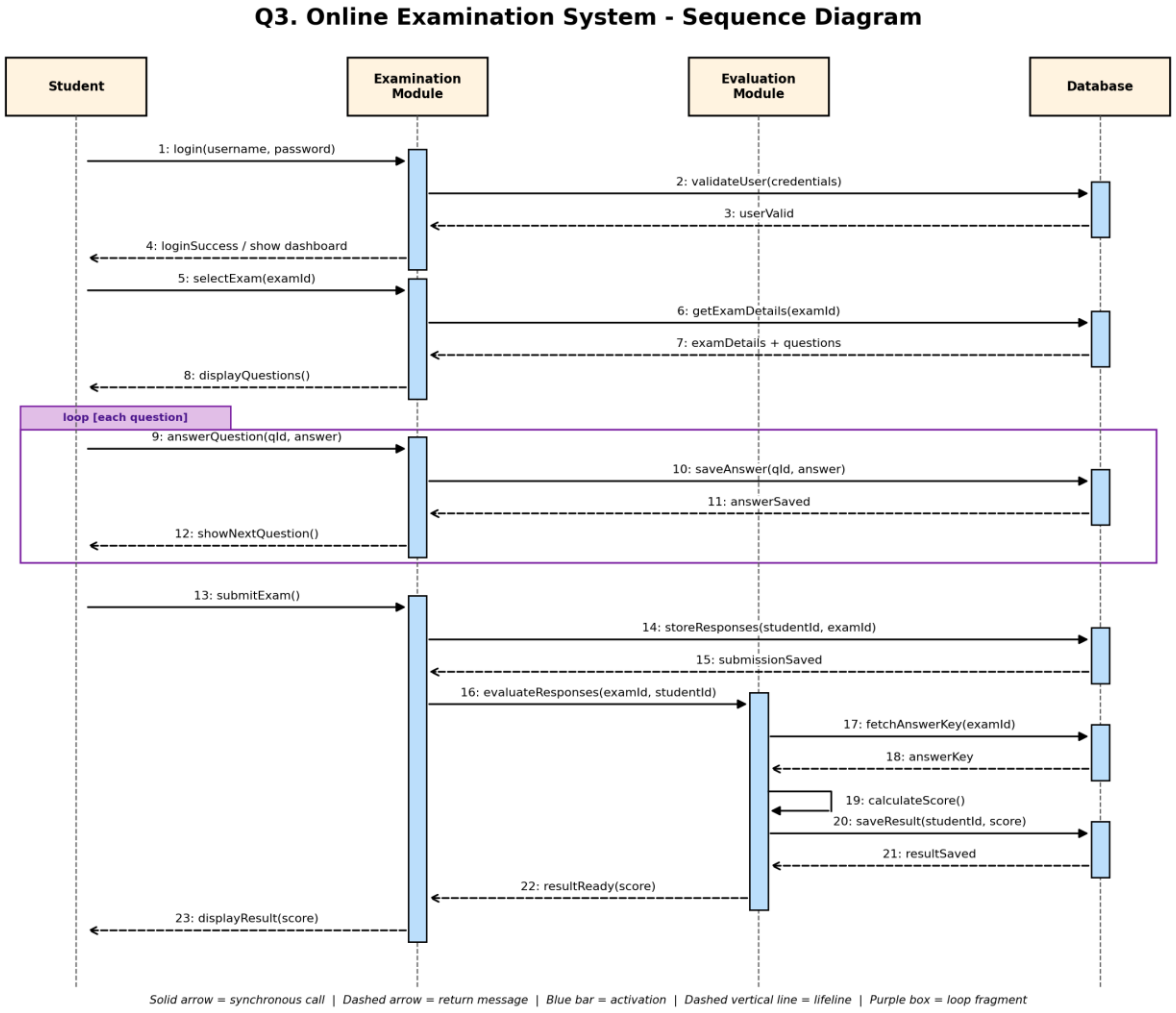


3. Sequence Diagram Evaluation

An **Online Examination System** enables students to log in, select an examination, answer questions, submit responses, and receive results. The communication among the student, examination module, evaluation module, and database is incomplete.

Analyze the interaction process, identify all participating objects and missing message exchanges, and design a detailed **Sequence Diagram** representing the complete interaction flow with activation bars and lifelines.

(10 Marks)

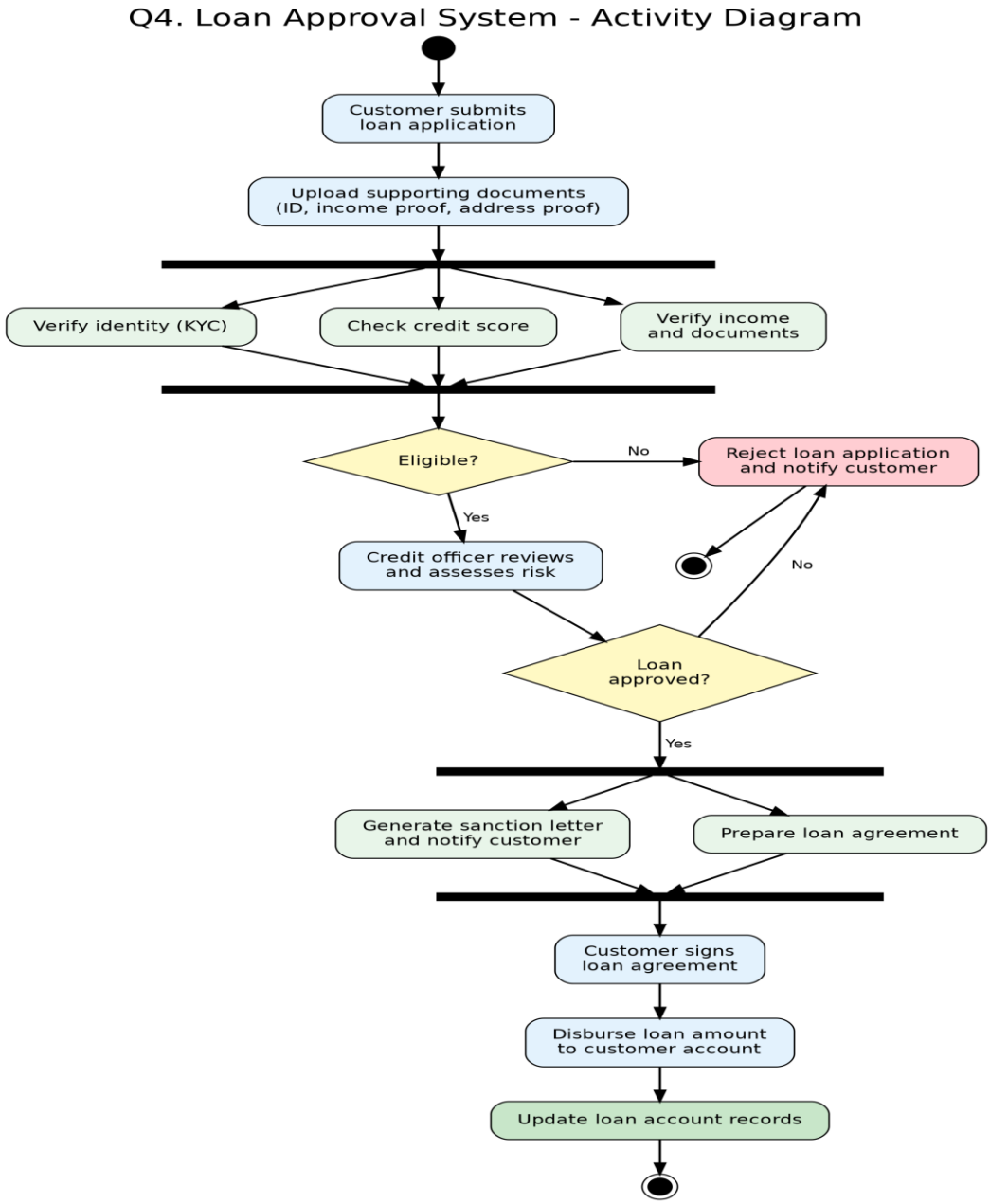


4. Activity Diagram Construction

A **Loan Approval System** allows customers to submit loan applications, upload supporting documents, complete eligibility verification, receive loan approval or rejection, and complete loan disbursement. The workflow lacks proper decision points and parallel activities.

Analyze the business process, identify all workflow activities, include decision nodes for approval and rejection, represent parallel processing where applicable, and construct an optimized **Activity Diagram**.

(10 Marks)

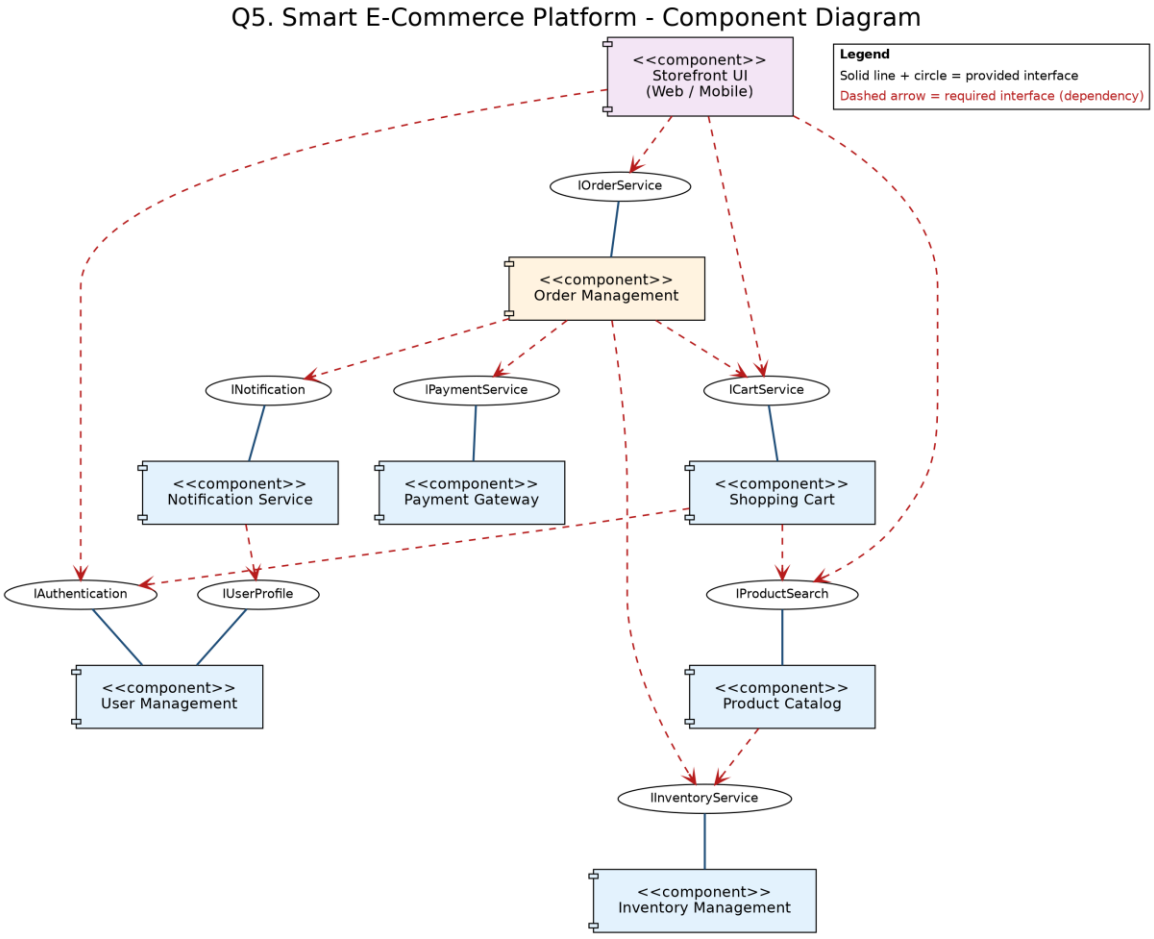


5. Component Diagram Design

A **Smart E-Commerce Platform** consists of modules such as User Management, Product Catalog, Shopping Cart, Order Management, Payment Gateway, Inventory Management, and Notification Service. The existing architecture does not clearly define interfaces and dependencies among modules.

Analyze the software architecture, identify major components and their provided/required interfaces, define dependencies among components, and design a comprehensive **Component Diagram** representing the overall system architecture.

(10 Marks)

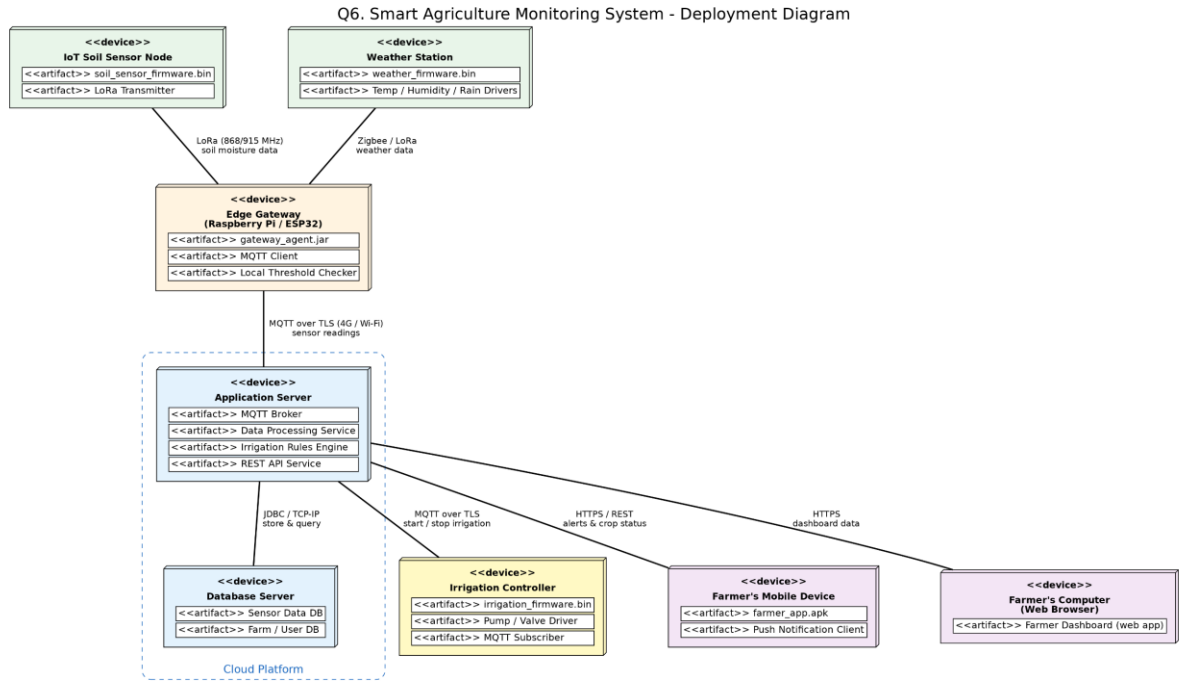


6. Deployment Diagram Design

A **Smart Agriculture Monitoring System** uses IoT soil sensors, weather stations, irrigation controllers, cloud servers, mobile applications, and farmer dashboards to monitor crop health and automate irrigation. The deployment architecture lacks proper representation of hardware nodes and communication links.

Analyze the deployment environment, identify hardware nodes and deployed software artifacts, specify communication protocols between nodes, and construct a detailed **Deployment Diagram** representing the complete physical system architecture.

(10 Marks)



RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand